

LOUIS WAN

chwanlouis@gmail.com · +852 63018527

EXPERIENCE

Agentic AI System for Intelligent Marketing & Consumer Insights

Personal Project

Jan 2026 - Present

- Engineered an autonomous agentic AI framework tailored for automated ad creative analysis, localized content generation, and intelligent consumer knowledge retrieval.
 - Optimized multimodal LLM performance for high-volume visual asset auditing by implementing an asset-splitting pipeline, achieving an **80% reduction in marketing campaign token costs**.
 - Developed a Knowledge Graph RAG pipeline to map brand guidelines and competitive intelligence, mitigating factual inconsistencies and lowering asset generation **hallucination rates by 70%**.
 - Built custom MCP servers to standardize real-time tool-calling, enabling seamless synchronization between creative LLMs and live marketing performance data sources.
-

CMU OmniClear (Hong Kong Monetary Authority)

Associate (Technology Service – Application Development)

Apr 2026 - Present

- Architecting backend solutions for **SIX Swiss Exchange** and **Bond Connect**, enabling seamless cross-border securities settlement and international custody services.
 - Developing secure microservices for workflows to facilitate multi-currency settlement and position reconciliation.
 - Designing complex data mappings and sequence diagrams to ensure high-availability connectivity for Northbound and Southbound trading links across global financial market infrastructures.
-

The Hongkong and Shanghai Banking Corporation Limited

Python developer, AI Market team, FICC, GBM

Jul 2024 - Dec 2025

- Designed, developed, and supported AI market, the flagship software for trading and analytics with Python, Java and kdb as tech stack
 - Collaborated with FX options and equity quant teams to implement analytical tools, enhance risk management capabilities and built MCP servers for training large language models within AI market
 - Optimized performance and scalability to handle high-frequency data and large-scale market data processing
 - Provided technical support and authored comprehensive documentation to ensure reliability and usability
-

GreySpark Partners

Developer

Projects included,

Hong Kong

Mar 2020 - JAN 2024

JEFFERIES

Python RAD developer, Quant strategy team

Jun 2023 - NOV 2023

- Developed and implemented an automated ETL solution for fundamental data streaming from FactSet, Bloomberg with reliability and scalability
 - Enhanced data visualization, providing clients with insights for informed decision-making
 - Engineered end-to-end automation for NASDAQ stocks, covering data streaming, cointegration model building, backtesting, and signal generation
 - Implemented a robust and scalable system that significantly reduced the time required for quantitative analysis, parameter tuning and strategy development
 - Actively contributed to Agile methodologies, managing projects efficiently through JIRA and maintaining documentation for transparency
-

HONG KONG SECURITIES AND FUTURES COMMISSION

Python developer, Machine learning engineer

Hong Kong

2021 Jun - 2023 Jun

- Worked as a python developer and machine learning engineer in data analytics project (DS-OL) with financial regulator.

- Developed machine learning algorithm and quantitative analysis for trade order data and FIX messages
- Successfully transformed 7 types of regulation into programming logic by python and SQL in Microsoft SQL server
- Assisted in web app developement and DS-OL data visualization

HAITONG INTERNATIONAL SECURITIES

Python and Java developer

Hong Kong
2020 Mar - 2021 Jun

- Served as a back-office Java and Oracle developer and implemented a data transformer handling trade and order data from Fidessa
- Worked as a python developer and finished Front Arena upgrade project and Apache Airflow project

Run Hua Investments Group Company Limited

Junior trader & Python developer

Hong Kong
Nov 2017 - Mar 2020

- Successfully handled Quant strategy based on CTA fund with over 20M HKD AUM
- Built a semi-automated trading system establish connectivity between Reuter's/Bloomberg data API and IB trading API
- Built software for data visualization of portfolio performance
- Developed new quantitative strategy and financial market research
- Assisted placing order for position changing and futures contract rollover with trading plan

Hong Kong Applied Science and Technology Research Institute

Python/ R Developer, Data analytics

Hong Kong
Jun 2015 - Nov 2017

- Participated in multiple projects in algo trading and financial modelling. Built portfolio and trading strategies in HK and US market with quantitative tools
- Successfully applied machine learning-based and optimization methods in various projects
- Assisted in deep learning research (Publication: S. S. Y. Ng ; W. Zhu ; W. W. S. Tang ; L. C. H. Wan ; A. Y. W. Wat 2016. An independent study of two deep learning platforms - H2O and SINGA. IEEE International Conference on Industrial Engineering and Engineering Management)

CERTIFICATION

Amazon web service

AWS Certified Cloud Practitioner (Validation number: G23NWM7C91FE1PCL)

2023 AUG

CQF institute by Fitch learning

Certificate in Quantitative Finance (CQF)

2022 JUN - 2023 MAR

Passed with Distinction

EDUCATION

DeepLearning.AI via Coursera

DeepLearning.AI TensorFlow Developer

2020

Udacity

Machine Learning Engineer Nanodegree

2017 - 2019

Hong Kong Polytechnic University

BSc(Hons) Investment science

2011 - 2015

SKILLS

Python:	NumPy, Pandas, Scikit-learn, Tensorflow, BeautifulSoup, Flask, asyncio, multiprocessing, blpapi
Java:	Spring boot framework
Database:	Microsoft SQL server, Oracle, PostgreSQL
Analytics tools:	Quantitative Finance, Machine Learning, Data Science, Statistical Modeling
Human Language:	native in Cantonese, fluent in English and Mandarin